Generating Functions, Field Theory, and Analytic Combinatorics

A Unified Tutorial on the Stochastic SIR Epidemic,
Doi–Peliti Reaction–Diffusion Theory,
and the Analytic Combinatorics of Critical Exponents

This document develops three parallel frameworks—probability generating functions for stochastic processes, Doi–Peliti field theory for reaction–diffusion systems, and Flajolet's analytic combinatorics—showing that all three share a common algebraic backbone and arrive at the same critical exponents by different routes. Every identity is motivated and every definition is given before it is used.

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1 Introduction and Motivation

Three apparently distinct bodies of mathematics meet at a single generating function.

- **Stochastic epidemiology** asks: given a random SIR epidemic, what is the probability of an outbreak of size n? The answer is encoded in a probability generating function (PGF) satisfying a first-order PDE whose *characteristics are the deterministic SIR orbits.*
- **Doi–Peliti field theory** asks: what is the long-time density of particles undergoing the reaction $A + A \to \emptyset$? The answer emerges from a path integral whose saddle-point equation is identical to the mean-field rate equation.
- **Analytic combinatorics** asks: given a recursively defined class of combinatorial objects, how many exist of size n, and what is the asymptotic growth? The answer comes from the type of singularity of the corresponding generating function.

The central message is:

The Unifying Principle

A phase transition, epidemic threshold, or asymptotic exponent is a **singularity** of a generating function. The *type* of singularity (pole, square-root branch point, logarithm) determines the universality class. The three frameworks are three different algebraic routes to the same singularity.

1.1 Notation and Conventions

Throughout this document:

- $[z^n]f(z)$ denotes the coefficient of z^n in the power series $f(z)$.
- $f(z) \sim g(z)$ as $z \to z_*$ means $f(z)/g(z) \to 1$.
- $a_n \sim C \cdot \alpha^n \cdot n^\beta$ means the ratio of the two sides tends to 1 as $n \to \infty$.
- $[a, b] = ab - ba$ is the commutator of operators a and b.
- $\Gamma(s)$ is the Euler gamma function, with $\Gamma(1/2) = \sqrt{\pi}$.
- All generating functions are *formal* power series unless the context specifies analytic continuation.

A quick-reference table of standard results, notation, and literature is provided in Appendix A.

2 Part I: The Stochastic SIR Epidemic

2.1 The Deterministic SIR System

The Kermack–McKendrick SIR model [1] partitions a population of size N into susceptible (S), infectious (I), and recovered (R) compartments. The deterministic dynamics are:

$$\frac{dS}{dt} = -\beta SI, \qquad \frac{dI}{dt} = \beta SI - \gamma I, \qquad \frac{dR}{dt} = \gamma I, \tag{1}$$

where $\beta > 0$ is the infection rate and $\gamma > 0$ is the recovery rate. The *basic reproduction number* is

$$R_0 = \frac{\beta N}{\gamma}. \tag{2}$$

2.1.1 The Conservation Law

Dividing the I-equation by the S-equation:

$$\frac{dI}{dS} = \frac{\beta SI - \gamma I}{-\beta SI} = -1 + \frac{\gamma}{\beta S} = -1 + \frac{1}{R_0 S/N}.$$

Integrating (with $s = S/N$ the normalised susceptible fraction):

$$\boxed{I + S - \frac{N}{R_0} \ln S = C_0,} \tag{3}$$

where $C_0 = I(0) + S(0) - \frac{N}{R_0} \ln S(0)$ is fixed by initial conditions. Equation (3) defines the **SIR orbit** in (S, I) space. It is the single most important equation in this document; we will rederive it from two completely different directions in Parts II and III.

2.1.2 The Epidemic Threshold

The epidemic peak occurs where $dI/dt = 0$, which gives $S_* = N/R_0$, i.e. the threshold susceptible fraction is $s_* = 1/R_0$. The epidemic grows if $S > S_*$ (i.e. $R_0 > 1$) and decays if $S < S_*$.

Epidemic Threshold

$R_0 = 1$ is the critical point. For $R_0 > 1$ a major epidemic is possible; for $R_0 < 1$ the epidemic always dies out. The bifurcation is continuous (second-order in the statistical-physics sense).

2.2 The Stochastic SIR: Master Equation

The stochastic SIR is a continuous-time Markov chain on (S, I) with transitions:

$$(S, I) \to (S - 1, I + 1) \quad \text{at rate} \quad \frac{\beta SI}{N}, \qquad (S, I) \to (S, I - 1) \quad \text{at rate} \quad \gamma I.$$

Let $P(s, i, t)$ be the probability of state $(S = s, I = i)$ at time t. The **master equation** (Kolmogorov forward equation) is:

$$\partial_t P(s,i,t) = \frac{\beta(s+1)(i-1)}{N} P(s+1,i-1,t) + \gamma(i+1)P(s,i+1,t) - \left[\frac{\beta si}{N} + \gamma i\right] P(s,i,t). \tag{4}$$

This is a system of $(N+1)^2$ coupled ODEs — impossible to solve directly for large N.

2.3 The Probability Generating Function

Definition 2.1 (Joint PGF)**.** The joint probability generating function of (S, I) is:

$$G(x,y,t) = \sum_{s=0}^{N} \sum_{i=0}^{N} P(s,i,t)\, x^s y^i. \tag{5}$$

The dummy variables x and y serve purely as *size markers*: the coefficient of $x^s y^i$ in G is the probability of exactly s susceptibles and i infectives. No probabilistic meaning is attached to x or y themselves.

2.3.1 Why This Helps

The shift operators in the master equation act as multiplication and differentiation on G. Specifically:

$$\sum_{s,i} s\, P(s,i) x^s y^i = x\, \partial_x G,$$
$$\sum_{s,i} (s+1) P(s+1,i-1) x^s y^i = \partial_x G \cdot \frac{y}{1} \cdot \ldots$$

After careful bookkeeping (multiplying each term in (4) by $x^s y^i$ and summing), the master equation becomes a **single PDE**:

$$\partial_t G = \frac{\beta}{N}(y-x)\, \partial_x(x\, \partial_y G) + \gamma(1-y)\, \partial_y G. \tag{6}$$

This is a first-order linear PDE in (x, y, t) — amenable to the method of characteristics.

2.4 Method of Characteristics

2.4.1 Setup

For a PDE of the form $\partial_t G + A(x,y)\partial_x G + B(x,y)\partial_y G = 0$, the **characteristics** are curves $(x(t), y(t))$ along which G is constant. They satisfy:

$$\frac{dx}{dt} = A(x,y), \qquad \frac{dy}{dt} = B(x,y).$$

Reading off from (6) (after rearranging to standard form):

$$\frac{dx}{dt} = -\frac{\beta}{N} x(y-1), \qquad \frac{dy}{dt} = \frac{\beta}{N} y(y-1) - \gamma(y-1) = (y-1)\left(\frac{\beta y}{N} - \gamma\right). \tag{7}$$

2.4.2 The Key Result

Theorem 2.2 (Kendall 1956, [3])**.** *The characteristic equations (7) of the SIR PGF PDE are, under the substitution $x = S/N$, $y = I/N$, identical to the deterministic SIR equations (1). Therefore, the probability distribution $P(s, i, t)$ is transported along the deterministic SIR orbits in (S, I) space.*

Verification. Under $x = S/N$, $\dot{x} = \dot{S}/N = -\beta SI/N^2 = -(\beta/N)x(Ni/N)$. Comparing with (7): $\dot{x} = -(\beta/N)x(y-1)$. At leading order in I/N, $y - 1 \approx y = I/N$, giving $\dot{x} = -(\beta/N)x \cdot I/N$, which matches $\dot{S}/N$. The y-equation similarly recovers $\dot{I}$. □ □

The Stochastic–Deterministic Bridge

The stochastic probability distribution $P(s, i, t)$ is constant along the *deterministic* SIR trajectories. The deterministic system is not an approximation to the stochastic one: they share the same invariant curves exactly, at the level of the generating function.

This result was first made explicit by Kendall [3] and placed in the PDE framework by Chalub and Souza [4]. Whittle [2] had earlier used generating function methods to derive the final-size distribution.

2.5 Critical Exponents for the SIR

2.5.1 Final Size Distribution

Solving along characteristics and applying boundary conditions at $I = 0$ gives the **final size distribution** (the probability that exactly n individuals were infected in total). For the Markovian SIR with one initial infective in a large population:

$$P(\text{total size} = n) = \frac{e^{-R_0 n}(R_0 n)^{n-1}}{n!}. \tag{8}$$

This is the **Borel distribution**. It is derivable directly from Lagrange inversion (see Part III). The generating function $T(z) = \sum_n P_n z^n$ satisfies:

$$T(z) = z\, e^{R_0(T(z)-1)}. \tag{9}$$

2.5.2 Singularity and Critical Exponent

The singularity of $T(z)$ occurs where the implicit function theorem fails (see Section 4.5 for the general theory). Setting $\partial/\partial T[T - ze^{R_0(T-1)}] = 0$:

$$T_* = \frac{1}{R_0}, \qquad z_* = \frac{1}{R_0} e^{1-1/R_0}. \tag{10}$$

Near z_*, Taylor expanding to second order:

$$T(z) \approx \frac{1}{R_0} - \sqrt{\frac{2}{R_0}} \cdot (z_* - z)^{1/2} + \mathcal{O}(z_* - z). \tag{11}$$

This is a **square-root branch point**. By the transfer theorem (Section **??**):

$$\boxed{P(\text{size} = n) \sim C \cdot n^{-3/2} \cdot z_*^{-n}, \qquad n \to \infty.} \tag{12}$$

At criticality $R_0 = 1$: $z_* = 1$, and the z_*^{-n} factor disappears, leaving the **pure power law** $P_n \sim Cn^{-3/2}$.

SIR Critical Exponent Summary

Quantity	Formula	Derivation
Epidemic threshold	$R_0 = 1$	$T_* = 1/R_0 \to 1$
GF singularity	$z_* = e^{1-1/R_0}/R_0$	IFT failure
Singularity type	Square-root branch	$F_{TT} \neq 0$
Final-size decay	$P_n \sim n^{-3/2}$ at $R_0 = 1$	Transfer theorem
Final-size GF	Borel distribution	Lagrange inversion

3 Part II: Doi–Peliti Field Theory

3.1 Physical Motivation

Consider a population of identical particles A in a volume, undergoing diffusion and the annihilation reaction $A + A \to \emptyset$ at rate λ. In one dimension, the mean particle density $\rho(t)$ is observed to decay as:

$$\rho(t) \sim t^{-1/2}, \qquad t \to \infty.$$

Deriving this rigorously — including the universal amplitude — requires going beyond the mean-field rate equation $\dot{\rho} = -\lambda\rho^2$, which gives only $\rho \sim (\lambda t)^{-1}$. Fluctuations dominate in low dimensions, and a field-theoretic treatment is needed.

3.2 Second Quantisation of the Master Equation

3.2.1 The Fock Space

Let the state of the system at site x be a non-negative integer n_x (the particle number). Define a **Fock space** $\mathcal{F}$ with basis vectors $|n\rangle$ (for a single site, suppressing spatial index for now). The state of the system at time t is encoded in the **probability vector**:

$$|\psi(t)\rangle = \sum_{n=0}^{\infty} P(n,t)\,|n\rangle. \tag{13}$$

Definition 3.1 (Creation and Annihilation Operators)**.** Define operators a (annihilation) and $a^\dagger$ (creation) by their action on basis states:

$$a\,|n\rangle = n\,|n-1\rangle, \qquad a^\dagger|n\rangle = |n+1\rangle. \tag{14}$$

These satisfy the canonical commutation relation:

$$\boxed{[a, a^\dagger] = 1.} \tag{15}$$

This is the **Weyl algebra** (or Heisenberg algebra), identical to the algebra of quantum mechanical ladder operators. The difference from quantum mechanics is purely interpretational: here $|n\rangle$ is a probability state, not a quantum state, so no Born rule applies.

The number operator $\hat{N} = a^\dagger a$ satisfies $\hat{N}|n\rangle = n|n\rangle$.

3.2.2 Why Annihilation and Creation?

In the master equation, the transition $n \to n-k$ (a reaction removing k particles) appears as:

$$\text{rate} \times P(n) \sim \text{rate} \times [n(n-1)\cdots(n-k+1)]\,P(n).$$

The prefactor $n(n-1)\cdots(n-k+1)$ is exactly the action of a^k on $|n\rangle$ (followed by appropriate factors). So the master equation becomes:

$$\partial_t|\psi\rangle = \mathcal{H}\,|\psi\rangle, \tag{16}$$

where $\mathcal{H}$ is a polynomial in a and $a^\dagger$ called the **Liouvillian** (or pseudo-Hamiltonian). This was Doi's key insight [5, 6].

3.2.3 The Liouvillian for $A + A \to \emptyset$

For pair annihilation at rate λ, the transition $n \to n-2$ occurs at rate $\lambda n(n-1)$. Writing the master equation and expressing it in terms of $a, a^\dagger$:

$$\mathcal{H} = \lambda\big(1 - a^{\dagger 2}\big)a^2. \tag{17}$$

Remark. *The structure (final state operator − initial state operator) × rate operator is universal: any reaction $kA \to \ell A$ gives $\mathcal{H} = r(a^{\dagger\ell} - a^{\dagger k})a^k$. See Appendix A.*

3.3 The Generating Function as a Coherent State

3.3.1 Representation on Power Series

Represent $a^\dagger \to z$ (multiplication by z) and $a \to \partial_z$ (differentiation). Then:

$$|\psi(t)\rangle \equiv G(z,t) = \sum_{n=0}^{\infty} P(n,t)\, z^n, \tag{18}$$

and the evolution equation (16) becomes:

$$\partial_t G(z,t) = \mathcal{H}(z,\partial_z)\, G(z,t). \tag{19}$$

The Fundamental Identification

The probability generating function $G(z,t) = \sum_n P(n,t)z^n$ is *identical* to the Doi Fock-space state vector in the representation $a^\dagger = z$, $a = \partial_z$. The Weyl algebra acts on the space of formal power series exactly as on Fock space. This is not an analogy: it is an isomorphism.

3.3.2 The Weyl Algebra Explicitly

The Weyl algebra W_1 is the algebra of polynomial differential operators generated by z and ∂_z with relation $[\partial_z, z] = 1$. It acts on $\mathbb{C}[[z]]$ (formal power series) via:

$$z \cdot \sum_n c_n z^n = \sum_n c_n z^{n+1}, \qquad \partial_z \cdot \sum_n c_n z^n = \sum_n n\, c_n z^{n-1}.$$

Every reaction Liouvillian is an element of W_1 (or its spatial version $W_1 \otimes C^\infty(x)$ for field theories). The master equation is an ODE in W_1.

3.4 The Path Integral (Peliti's Construction)

Peliti [7] showed how to convert the operator evolution (16) into a functional integral by inserting resolutions of identity in coherent states at each time step. The result is:

$$\langle \text{observable} \rangle = \int \mathcal{D}[\phi, \tilde{\phi}]\; \mathcal{O}[\phi]\; e^{-S[\phi,\tilde{\phi}]}, \tag{20}$$

where the **action** for a single-species system with reaction $A + A \to \emptyset$ and diffusion is (after the *Doi shift* $\tilde{\phi} \to 1 + \tilde{\phi}$):

$$S = \int d^d x\, dt\; \left[\tilde{\phi}\,(\partial_t - D\nabla^2)\phi + 2\lambda\,\tilde{\phi}\phi^2 + \lambda\,\tilde{\phi}^2\phi^2\right]. \tag{21}$$

The fields $\phi(x,t)$ and $\tilde{\phi}(x,t)$ are the Peliti *response field* and *density field* respectively. The physical density is $\rho(x,t) = \langle\phi(x,t)\rangle$.

3.4.1 Mean-Field Limit

Ignoring fluctuations ($\tilde{\phi} \to 0$, saddle-point approximation) and setting $\phi = \rho$ uniform: only the $\tilde{\phi}\phi^2$ term survives in the variation, giving $\partial_t\rho = -2\lambda\rho^2$, i.e. the mean-field rate equation. The path integral saddle is the deterministic orbit, exactly as the PGF characteristic is the SIR orbit.

3.5 Renormalisation Group Analysis

3.5.1 The Ward Identity Cancellation

For the reaction $A + A \to \emptyset$ in d dimensions, the quartic vertex $\tilde{\phi}^2\phi^2$ in (21) is exactly cancelled by a counter-term generated by the cubic vertex $\tilde{\phi}\phi^2$ at one loop. This follows from a Ward identity associated with the probability conservation constraint $H(1,a) = 0$ (see [9] for the proof). The effective action is:

$$S_{\text{eff}} = \int d^d x\, dt\, \tilde{\phi}\,(-\partial_t + D\nabla^2)\phi, \tag{22}$$

i.e. the theory is *Gaussian* in the IR. This is an exact statement, valid to all loop orders in $d \leq 2$.

3.5.2 Computing the Density Decay

From (22), the retarded Green's function (propagator) is:

$$G_R(k,\omega) = \frac{1}{-i\omega + Dk^2}. \tag{23}$$

The density at time t is obtained by integrating out all wavenumber modes:

$$\rho(t) = \rho_0 \int \frac{d^d k}{(2\pi)^d}\, e^{-Dk^2 t} = \frac{\rho_0}{(4\pi D t)^{d/2}}. \tag{24}$$

In $d = 1$:

$$\boxed{\rho(t) \sim t^{-1/2}.} \tag{25}$$

The exponent 1/2 comes from the Gaussian integral over a single momentum component in one spatial dimension. The upper critical dimension is $d_c = 2$; for $d > d_c$ the mean-field result $\rho \sim t^{-1}$ holds.

3.5.3 The Full Doi–Peliti Calculation: Lee (1994)

The above treatment is schematic. The complete RG calculation for $kA \to \emptyset$ was performed by Lee [8], who showed:

- The density decay exponent $\alpha_d = d/2$ for $d < d_c = 2/(k-1)$ is *exact* — determined by dimensional analysis alone, not corrected by loops.
- For $k = 2$ (pair annihilation), $d_c = 2$ and $\alpha_1 = 1/2$ is exact.
- The amplitude requires an explicit one-loop calculation.

The comprehensive review by Täuber, Howard, and Vollmayr-Lee [9] presents the full pedagogical derivation including Feynman rules, renormalisation, and the Callan–Symanzik equation.

Doi–Peliti Critical Exponent Summary

Quantity	Formula	Source
Upper critical dimension	$d_c = 2/(k-1)$	Dimensional analysis
Density decay ($d < d_c$)	$\rho \sim t^{-d/2}$	Gaussian integral
Density decay ($d > d_c$)	$\rho \sim t^{-1}$	Mean-field
Density decay ($d = 1$, $k = 2$)	$\rho \sim t^{-1/2}$	Exact
Exponent exactness	All orders in $\epsilon = d_c - d$	Ward identity

3.6 The Linking Structure: Same Square Root

Before proceeding to analytic combinatorics, it is worth noting the common algebraic structure between the SIR and Doi–Peliti results.

In Laplace space ($t \to s$), the density propagator is:

$$\hat{\rho}(s) = \int \frac{d^d k}{(2\pi)^d} \frac{1}{s + Dk^2} \sim s^{-1+d/2} \qquad (d = 1) \implies \hat{\rho}(s) \sim s^{-1/2}.$$

For the SIR final-size distribution, the Laplace transform of the $n^{-3/2}$ tail is:

$$\sum_n n^{-3/2} e^{-sn} \sim \int_0^\infty n^{-3/2} e^{-sn}\, \mathrm{d}n = \Gamma(-1/2) \cdot s^{1/2} \sim s^{1/2}.$$

The survival probability $P_{\text{surv}}(t) = \sum_{n>t} P_n \sim t^{-1/2}$ in Laplace space gives $\sim s^{-1/2}$.

Both are shadows of the same $\sqrt{s}$, which is the Laplace transform signature of Brownian motion. The square root appears because the underlying conserved quantity — the SIR orbit (3) and the Doi diffusion propagator (23) — both encode a one-dimensional Gaussian random walk.

4 Part III: Analytic Combinatorics

4.1 What is Analytic Combinatorics?

Analytic combinatorics [11] is a systematic framework for translating between:

1. The **recursive structure** of a combinatorial class (how its objects are built from smaller objects), and
2. The **asymptotic count** of objects of size n.

The bridge is the generating function. The route is:

$$\textbf{Grammar} \xrightarrow{\text{symbolic method}} \textbf{GF equation} \xrightarrow{\text{complex analysis}} \textbf{Asymptotics.}$$

The power of the method is that both arrows are *mechanical*: given the grammar, the GF equation follows by a dictionary; given the GF equation, asymptotics follow from the type of its dominant singularity.

4.2 Combinatorial Classes and Generating Functions

Definition 4.1 (Combinatorial Class)**.** A **combinatorial class** is a set $\mathcal{A}$ with a *size* function $|\cdot| : \mathcal{A} \to \mathbb{N}$ such that the number of objects of each size is finite. Write $a_n = |\{\alpha \in \mathcal{A} : |\alpha| = n\}|$.

Definition 4.2 (Ordinary Generating Function)**.** The **ordinary generating function (OGF)** of a class $\mathcal{A}$ is the formal power series:

$$A(z) = \sum_{n \geq 0} a_n z^n. \tag{26}$$

The variable z is a *size marker*: the coefficient of z^n counts objects of size n.

Definition 4.3 (Exponential Generating Function)**.** For *labelled* structures (where atoms are distinguishable), use the **exponential generating function (EGF)**:

$$\hat{A}(z) = \sum_{n \geq 0} a_n \frac{z^n}{n!}. \tag{27}$$

4.2.1 Why Two Kinds?

In an unlabelled structure, atoms are indistinguishable (e.g. binary trees with identical leaves). In a labelled structure, each atom carries a unique label $1, \ldots, n$ (e.g. permutations, Poisson-offspring trees).

The factor $1/n!$ in the EGF precisely cancels the overcounting from relabellings. The combinatorial constructions of Section 4.3 look identical for both, but with different GFs.

4.3 The Symbolic Method

The key insight is that set-theoretic operations on combinatorial classes correspond to algebraic operations on generating functions. The following table holds for both OGFs (unlabelled) and EGFs (labelled):

Construction	Notation	OGF	EGF
Disjoint union	$\mathcal{A}+\mathcal{B}$	$A(z)+B(z)$	$\hat{A}(z)+\hat{B}(z)$
Cartesian product	$\mathcal{A}\times\mathcal{B}$	$A(z)\cdot B(z)$	$\hat{A}(z)\cdot\hat{B}(z)$
Sequence	$\mathrm{SEQ}(\mathcal{A})$	$\frac{1}{1-A(z)}$	$\frac{1}{1-\hat{A}(z)}$
Set (unlabelled)	$\mathrm{MSET}(\mathcal{A})$	$\exp\sum_{k\geq 1}\frac{A(z^k)}{k}$	—
Set (labelled)	$\mathrm{SET}(\mathcal{A})$	—	$e^{\hat{A}(z)}$
Cycle	$\mathrm{CYC}(\mathcal{A})$	$\log\frac{1}{1-A(z)}$	$\log\frac{1}{1-\hat{A}(z)}$
Pointing	$\Theta\mathcal{A}$	$z\,A'(z)$	$z\,\hat{A}'(z)$

4.3.1 Why These Rules Hold

Disjoint union: The a_n+b_n objects of size n in $\mathcal{A}+\mathcal{B}$ contribute $a_nz^n+b_nz^n=(a_n+b_n)z^n$.

Cartesian product: An object in $\mathcal{A}\times\mathcal{B}$ of size n is a pair (α,β) with $|\alpha|+|\beta|=n$. Summing over all splits: $\sum_{j=0}^{n}a_jb_{n-j}$, which is the coefficient of z^n in $A(z)B(z)$. This is just the Cauchy product of series.

Sequence: $\mathrm{SEQ}(\mathcal{A})=\epsilon+\mathcal{A}+\mathcal{A}^2+\ldots$ where ϵ is the empty sequence. Translating: $1+A+A^2+\ldots=1/(1-A)$.

Labelled set: A set of k labelled objects from $\mathcal{A}$ has EGF $\hat{A}(z)^k/k!$ (divide by $k!$ because the set has no order). Summing over k: $\sum_{k\geq 0}\hat{A}^k/k!=e^{\hat{A}(z)}$.

Example 4.4 (Binary Trees)**.** A binary tree $\mathcal{T}$ is either a leaf $\bullet$ (size 1) or a pair of binary trees joined at a root (size = 1+ size of left subtree + size of right subtree). The grammar is:

$$\mathcal{T}=\{\bullet\}+\bullet\times\mathcal{T}\times\mathcal{T}.$$

Translating: $T(z)=z+z\cdot T(z)^2$. Solving: $T(z)=(1-\sqrt{1-4z})/2$. Expanding: $[z^n]T(z)=C_{n-1}$ where $C_n=\binom{2n}{n}/(n+1)$ is the n-th Catalan number.

4.4 Singularity Analysis

4.4.1 Why Singularities Determine Asymptotics

The coefficients of a GF are recovered by Cauchy's integral formula:

$$a_n=[z^n]\,A(z)=\frac{1}{2\pi i}\oint_{|z|=r}\frac{A(z)}{z^{n+1}}\,\mathrm{d}z. \tag{28}$$

For large n, this contour integral is dominated by the **singularities of $A(z)$ closest to the origin** (smallest $|z_*|$). All the large-n information is compressed into the nature of these singularities.

4.4.2 The Transfer Theorems

Let $\rho=1/|z_*|$ be the exponential growth rate and write $A(z)\sim K(1-z/z_*)^\alpha$ near z_*.

Theorem 4.5 (Transfer Theorem, Flajolet–Odlyzko [11])**.** *If $A(z)$ is analytic in a* dented domain *around z_* (see Fig. below) and $A(z)\sim K(1-z/z_*)^\alpha$ as $z\to z_*$ with $\alpha\notin\mathbb{N}$, then:*

$$[z^n]\,A(z)\sim\frac{K}{\Gamma(-\alpha)}\cdot n^{-\alpha-1}\cdot z_*^{-n}. \tag{29}$$

The key cases:

Singularity type	Local form	a_n asymptotics
Simple pole	$(1 - z/z_*)^{-1}$	$C \cdot z_*^{-n}$
k-th order pole	$(1 - z/z_*)^{-k}$	$C \cdot n^{k-1} \cdot z_*^{-n}$
Square-root branch	$(1 - z/z_*)^{1/2}$	$C \cdot n^{-3/2} \cdot z_*^{-n}$
Inverse square-root	$(1 - z/z_*)^{-1/2}$	$C \cdot n^{-1/2} \cdot z_*^{-n}$
Logarithm	$\log(1 - z/z_*)$	$C \cdot n^{-1} \cdot z_*^{-n}$
General α	$(1 - z/z_*)^{\alpha}$	$C \cdot n^{-\alpha-1} \cdot z_*^{-n}$

The $\Gamma(-\alpha)$ in (29) is the normalisation from the generalised binomial expansion: $(1-u)^\alpha = \sum_n \binom{\alpha}{n}(-u)^n$ and $\binom{\alpha}{n} \sim n^{-\alpha-1}/\Gamma(-\alpha)$.

4.4.3 The Dented Domain (Briefly)

The transfer theorem requires that $A(z)$ can be analytically continued to a region Δ that *dents into* the disk $|z| < |z_*|$ from outside the singularity. This excludes cases where there are infinitely many singularities on the circle $|z| = |z_*|$ (which can happen for combinatorial GFs with periodic structure). For all the GFs in this document, the condition is satisfied.

4.5 Lagrange Inversion

4.5.1 The Problem

Many combinatorial recursions take the **Lagrange form**:

$$T(z) = z\,\phi(T(z)), \tag{30}$$

for some known function ϕ. The GF $T(z)$ is implicitly defined and generally has no closed form. Lagrange (1770) gave a direct formula for its coefficients.

4.5.2 Origin: Fixed-Point Inversion

Equation (30) says that T is a fixed point of the map $T \mapsto z\phi(T)$. For small z, this map is a contraction and the fixed point is unique and analytic. As z increases, the map eventually ceases to be a contraction — this is exactly the singularity.

More precisely, the implicit function theorem guarantees that $T(z)$ is analytic near $z = 0$ as long as $\partial/\partial T[T - z\phi(T)] \neq 0$, i.e. as long as $1 \neq z\phi'(T)$.

Theorem 4.6 (Lagrange Inversion Formula)**.** *If $T = z\phi(T)$ and $\phi(0) \neq 0$, then:*

$$[z^n]\,T(z) = \frac{1}{n}\,[T^{n-1}]\,\phi(T)^n. \tag{31}$$

Proof sketch. Write $z = T/\phi(T)$ so that $dz = (1/\phi - T\phi'/\phi^2)dT$. By Cauchy's formula:

$$[z^n]T = \frac{1}{2\pi i}\oint \frac{T}{z^{n+1}}\,\mathrm{d}z = \frac{1}{2\pi i}\oint \frac{T}{(T/\phi(T))^{n+1}}\frac{dz}{dT}\,\mathrm{d}T.$$

Computing dz/dT, using residue calculus at $T = 0$, and simplifying yields (31). □ □

4.5.3 The Singularity of the Lagrange Equation

Setting $\partial F/\partial T = 0$ where $F(T, z) = T - z\phi(T)$:

$$1 = z_* \phi'(T_*), \qquad T_* = z_* \phi(T_*). \tag{32}$$

These two equations determine the branch point (T_*, z_*). Near (T_*, z_*), expanding to second order in $\delta T = T - T_*$:

$$0 = F_z(z_* - z) + \tfrac{1}{2} F_{TT} \, (\delta T)^2 + \dots$$

Since $F_z = -\phi(T_*)$ and $F_{TT} = -z_* \phi''(T_*)$:

$$\delta T \sim \sqrt{\frac{2\phi(T_*)}{z_* \phi''(T_*)}} \cdot (z_* - z)^{1/2}. \tag{33}$$

Any Lagrange equation with $\phi'' \neq 0$ at the critical point produces a square-root branch point. The exponent $-3/2$ in the coefficient asymptotics (via the transfer theorem) is therefore *universal* for this class of equations.

5 Part IV: Analytic Combinatorics Recovers Both Results

5.1 AC Derivation of the SIR Final-Size Distribution

5.1.1 The Epidemic as a Labelled Tree

In the large-N limit, each infectious individual independently infects a Poisson(R_0) number of others before recovering. Because each new infection is of a (hitherto) susceptible individual chosen uniformly at random from a large pool, infections are approximately independent — this is the *branching process approximation.*

The epidemic is therefore a **Galton–Watson tree** with Poisson(R_0) offspring. The total number of infected individuals (epidemic size) is the **total progeny** of this tree.

The symbolic description is: an epidemic tree $\mathcal{T}$ consists of a labelled root followed by a labelled set of sub-epidemics (one per infection):

$$\mathcal{T} = \bullet \times \mathrm{SET}(\mathcal{T}) \quad \text{(weighted by Poisson).} \tag{34}$$

In EGF language, with the Poisson weight $e^{R_0(T-1)}$:

$$T(z) = z\, e^{R_0(T(z)-1)}. \tag{35}$$

This is the Lagrange equation (30) with $\phi(T) = e^{R_0(T-1)}$.

5.1.2 Applying Lagrange Inversion

With $\phi(T) = e^{R_0(T-1)}$:

$$[z^n]T = \frac{1}{n}[T^{n-1}]\, e^{R_0 n(T-1)} = \frac{e^{-R_0 n}}{n} \cdot \frac{(R_0 n)^{n-1}}{(n-1)!} = \frac{e^{-R_0 n}(R_0 n)^{n-1}}{n!}.$$

This recovers the Borel distribution (8) in one line.

5.1.3 Finding the Singularity via the Lagrange Conditions

Applying (32) with $\phi(T) = e^{R_0(T-1)}$, $\phi'(T) = R_0 e^{R_0(T-1)}$:

$$1 = z_* R_0 e^{R_0(T_*-1)} \tag{36}$$

$$T_* = z_* e^{R_0(T_*-1)}. \tag{37}$$

Dividing: $T_* R_0 = 1$, giving $T_* = 1/R_0$. Substituting back: $z_* = (1/R_0)e^{R_0(1/R_0-1)} = (1/R_0)e^{1-1/R_0}$, confirming (10).

5.1.4 The Singular Locus Is the SIR Orbit

At the critical point, take logarithms of (37):

$$\ln T_* = \ln z_* + R_0(T_* - 1).$$

Rearranging:

$$\ln z_* = \ln T_* - R_0 T_* + R_0. \tag{38}$$

Now let T_* vary continuously (treating the singularity as a function of the critical point). Setting $T = S/N$ (susceptible fraction):

$$\ln z = \ln \frac{S}{N} - R_0 \frac{S}{N} + \text{const.} \tag{39}$$

Compare with the SIR conservation law (3): $I + S - (N/R_0) \ln S = C_0$, which rearranges to:

$$\ln S - R_0 \frac{S}{N} = \tilde{C}. \tag{40}$$

These are identical. The singular locus of the GF $T(z)$ traces the SIR orbit in $(S/N, z)$ space.

AC Recovers the SIR Conserved Quantity

The SIR conservation law $I + S - (N/R_0) \ln S = C$ is not an input to the analytic combinatorics calculation. It emerges as the **singular locus of the epidemic-tree generating function** under the substitution $T = S/N$. The method of characteristics (Part I) and the Lagrange singularity condition (Part III) are two different derivations of the same algebraic curve.

5.1.5 The Two Branches = Minor vs Major Epidemic

Near z_*, (33) gives two branches $T = T_* \pm C\sqrt{z_* - z}$:

- **Small branch** ($T < T_* = 1/R_0$): the epidemic is small, dies out before crossing the peak susceptible threshold. This is the *minor outbreak*.
- **Large branch** ($T > T_* = 1/R_0$): the epidemic crosses the peak and infects $O(N)$ individuals. This is the *major epidemic*.

The extinction probability q (probability of minor outbreak) is the value of $T(1)$ on the small branch: it satisfies $q = e^{R_0(q-1)}$, the classic Galton–Watson extinction equation.

5.2 AC Derivation of the $A + A \to \emptyset$ Exponent

5.2.1 Reduction to First Passage

In one spatial dimension, particles cannot cross each other. Label them $1, 2, \ldots$ from left to right. Particle k is annihilated when it meets particle $k+1$ (or $k-1$). In *relative* coordinates, the pair $(k, k+1)$ is described by a single random walk $X_t = x_{k+1}(t) - x_k(t)$ on the positive integers, starting at some $d > 0$. The pair annihilates when X_t first hits 0.

The density $\rho(t)$ decays because pairs annihilate; the annihilation rate is the first-passage rate of the relative random walk to 0. Specifically:

$$\rho(t) \sim P(\text{relative walk has not hit 0 by time } t) = P_{\text{surv}}(t). \tag{41}$$

5.2.2 The First-Passage GF

A **first-passage path of length** n from $d > 0$ to 0 is a nearest-neighbour path that stays strictly positive for steps $1, \ldots, n-1$ and reaches 0 at step n.

For the *symmetric* random walk (equal left/right probabilities $p = 1/2$), the GF for first-passage paths starting at $d = 1$ satisfies the functional equation:

$$F(z) = \frac{z}{2} + \frac{z}{2} \cdot F(z)^2, \tag{42}$$

since each path either takes one step directly to 0 (weight $z/2$), or takes one step right, executes a complete excursion back to 1, and then completes a first passage from 1 to 0 (weight $(z/2) \cdot F(z)^2$).

Remark. *Equation (42) is a Lagrange equation! With $p = 1/2$ and $F = z\phi(F)$ it reads $F(z) = z \cdot \frac{1+F^2}{2}$, so $\phi(F) = (1+F^2)/2$.*

Solving the quadratic:

$$F(z) = \frac{1 - \sqrt{1-z^2}}{z}. \tag{43}$$

5.2.3 The Singularity

$F(z)$ has a square-root branch point at $z_* = 1$ (where $1 - z^2 = 0$). Near $z = 1$:

$$F(z) \approx 1 - \sqrt{2(1-z)} + \mathcal{O}(1-z), \tag{44}$$

i.e. $F(z) \sim 1 - \sqrt{2} \cdot (1-z)^{1/2}$. With $\alpha = 1/2$, the transfer theorem (29) gives:

$$f_n = [z^n]F(z) \sim \frac{\sqrt{2}}{2} \cdot \frac{1}{\Gamma(-1/2)} \cdot n^{-3/2} = \frac{1}{\sqrt{2\pi}} \cdot n^{-3/2}. \tag{45}$$

(Using $\Gamma(-1/2) = -2\sqrt{\pi}$.)

5.2.4 Integrating to the Survival Probability

The survival probability is the complementary CDF of the first-passage distribution:

$$P_{\text{surv}}(t) = \sum_{n>t} f_n \sim \int_t^\infty \frac{dn}{\sqrt{2\pi}\, n^{3/2}} = \sqrt{\frac{2}{\pi t}}. \tag{46}$$

Therefore, from (41):

$$\boxed{\rho(t) \sim \left(\frac{2}{\pi t}\right)^{1/2} \sim t^{-1/2},} \tag{47}$$

in exact agreement with the Doi–Peliti result (25).

AC Recovers the Doi–Peliti Exponent

The exponent $-1/2$ comes from integrating the first-passage tail $f_n \sim n^{-3/2}$. The $n^{-3/2}$ comes from the transfer theorem applied to the square-root branch point of $F(z)$ at $z_* = 1$. The square-root branch point is forced by the Lagrange structure of the first-passage recursion. The chain is:

Annihilation $\to$ First-passage problem $\to$ Lagrange GF $\to$ Square-root branch $\to n^{-3/2} \to \rho(t) \sim t^{-1/2}$.

5.3 The Common Algebraic Root

Both derivations hinge on a **square-root branch point** in a GF satisfying a Lagrange equation. This is not a coincidence. The Lagrange equation encodes *branching structure*; and branching structure appears in both settings because:

- In the SIR, the epidemic is a Galton–Watson branching tree.
- In the annihilation problem, the first-passage path is built from excursions which are themselves branching trees (Motzkin paths / ballot paths).

In both cases, the *critical point* is where the average branching number equals one: $R_0 = 1$ for SIR, and the random walk at criticality (equal left/right probabilities) for first passage. At this criticality, the Lagrange singularity sits on the boundary of the unit disk ($z_* = 1$), and the asymptotics are pure power laws without exponential factors.

The Doi–Peliti calculation arrives at the same $t^{-1/2}$ via a completely different path: Gaussian integration over diffusion modes in momentum space. The Laplace-space correspondence

$$\int \frac{d^1k}{2\pi} \frac{1}{s + Dk^2} \sim s^{-1/2} \quad \longleftrightarrow \quad \sum_n n^{-3/2} e^{-sn} \sim s^{1/2}$$

identifies the two computations as Fourier/Laplace duals. The combinatorial $n^{-3/2}$ and the field-theoretic k^{-1} momentum integral are the same square root in different basis representations.

6 Part V: Analytic Combinatorics and Feynman Diagrams

The Thesis of This Section

The connection between AC and Feynman diagrams is not analogical. It is a literal identification of the same algebraic structures appearing simultaneously in combinatorics and quantum field theory. There are four distinct levels at which this identification holds, ranging from classical folklore to active research. We develop each in turn, with explicit examples, and close with an assessment of what is established and what remains open.

6.1 Connection 1: The Exponential Formula and $\log Z$

6.1.1 The Physics Statement

In any quantum field theory or statistical mechanics model, the partition function Z receives contributions from all Feynman diagrams (connected or not). The **free energy** $F = -\log Z$ receives contributions only from *connected* diagrams. This is used in every QFT textbook to discard disconnected vacuum bubbles.

6.1.2 The AC Statement

In analytic combinatorics, if $\mathcal{C}$ is a labelled class of *connected* structures, and $\mathcal{A} = \mathrm{SET}(\mathcal{C})$ is the class of all structures (each being a set of connected components), then:

$$\hat{A}(z) = e^{\hat{C}(z)}, \qquad \hat{C}(z) = \log \hat{A}(z). \tag{48}$$

This is the **exponential formula**. It follows immediately from the EGF rule for the SET construction: $\mathrm{SET}(\mathcal{C})$ translates to $e^{\hat{C}(z)}$.

6.1.3 They Are the Same Identity

The statement "$\log Z$ generates connected diagrams" is *exactly* (48) with:

$$\begin{aligned} \hat{A}(z) &\leftrightarrow Z \quad \text{(all diagrams, vacuum normalised)}, \\ \hat{C}(z) &\leftrightarrow \log Z \quad \text{(connected diagrams only)}, \\ z^n/n! &\leftrightarrow g^n \quad \text{(perturbative expansion in coupling } g\text{)}. \end{aligned}$$

The *symmetry factors* of Feynman diagrams — those awkward denominators $1/|\mathrm{Aut}(\Gamma)|$ that appear in loop calculations — are exactly the $1/n!$ overcounting factors in the EGF SET construction. They have a purely combinatorial origin: the automorphism group of a diagram is the symmetry group that acts on its legs and propagators, which is exactly the symmetry group counted by the EGF.

Example 6.1 (Symmetry Factor from the SET Construction)**.** A diagram Γ consisting of two identical one-loop bubbles (a "figure-eight" vacuum graph) has $|\mathrm{Aut}(\Gamma)| = 8$ in ϕ^4 theory. In EGF language:

$$\hat{C}_{\text{two-bubble}}(z) = \frac{1}{2!}\hat{C}_{\text{one-bubble}}(z)^2 = \frac{1}{2}\hat{C}_{\text{one-bubble}}(z)^2.$$

The $1/2!$ is the SET factor for a set of two identical objects. More generally, for a vacuum graph consisting of k copies of the same connected diagram γ:

$$\text{symmetry factor} = |\mathrm{Aut}(\gamma)|^k \cdot k!,$$

where $k!$ is from the EGF SET construction and $|\mathrm{Aut}(\gamma)|^k$ accounts for the internal symmetries of each copy.

Status

Completely classical. Appears in Flajolet & Sedgewick [11] as a standard EGF application, and in every QFT textbook (e.g. Zinn-Justin [21]) as the cluster expansion / cumulant theorem. The explicit identification of symmetry factors with EGF overcounting is noted in Cvitanović, Lautrup & Pearson (1978) and in Yeats [16].

6.2 Connection 2: Dyson–Schwinger Equations Are Lagrange Equations

6.2.1 The Dyson–Schwinger Equation

In ϕ^3 theory, the full (dressed) propagator $G(p^2)$ satisfies the Dyson–Schwinger equation:

$$G = G_0 + G_0 \cdot \Sigma(G) \cdot G, \tag{49}$$

where $G_0 = 1/(p^2 + m^2)$ is the bare propagator and $\Sigma(G)$ is the self-energy, which is itself a function of G (all 1PI insertions). Rearranging:

$$G = G_0\big[1 + \Sigma(G) \cdot G\big] \equiv G_0 \cdot \Phi(G). \tag{50}$$

This is exactly a **Lagrange equation** $T = z\phi(T)$ with:

$$T \leftrightarrow G, \quad z \leftrightarrow G_0, \quad \phi(T) = 1 + \Sigma(T) \cdot T.$$

6.2.2 Lagrange Inversion Generates the Perturbative Expansion

Applying (31):

$$[G_0^n]\, G = \frac{1}{n}\, [G^{n-1}]\, \Phi(G)^n. \tag{51}$$

Each term $[G^{n-1}]\Phi(G)^n$ counts the number of Feynman diagrams with n propagators, weighted by their amplitudes. The Lagrange inversion formula is therefore a **generating function for the perturbative expansion** of the dressed propagator.

6.2.3 The Singularity is the Landau Pole

By the Lagrange singularity conditions (32):

$$1 = G_0^* \Phi'(G^*), \qquad G^* = G_0^* \Phi(G^*).$$

The singularity of G (as a function of G_0 or equivalently of the coupling λ) is the **Landau pole**: the scale at which the perturbative expansion breaks down and non-perturbative effects take over. In AC terms, it is the branch point of the Lagrange GF. The type of singularity (square root, pole, essential) determines the non-perturbative physics.

Example 6.2 (Running Coupling in QED)**.** In QED, the running coupling $\alpha(\mu^2)$ satisfies a Dyson–Schwinger-type equation that, in the leading-log approximation, reduces to:

$$\alpha(\mu^2) = \frac{\alpha_0}{1 - (\alpha_0/3\pi)\log(\mu^2/m^2)}.$$

This has a simple pole at $\mu^2 = m^2 e^{3\pi/\alpha_0}$ — the Landau pole. In Lagrange language, $\phi(T) = 1/(1 - cT)$ produces a simple pole at $T_* = 1/c$, giving $z_* = 1/(ce)$ and $G \sim (G_0^* - G_0)^{-1}$ near the singularity. The transfer theorem then gives $[G_0^n]G \sim C \cdot (G_0^*)^{-n}$ — pure exponential growth in loop order, with no power-law correction.

Status

The identification of Dyson–Schwinger equations with fixed-point / Lagrange equations is known in the Connes–Kreimer community (Foissy [17], Yeats [16]). The explicit phrasing in terms of the AC Lagrange inversion theorem with the IFT singularity condition and transfer theorem is not, as far as we are aware, stated cleanly in the literature. The two communities (AC and QFT) use different language for the same object.

6.3 Connection 3: The Connes–Kreimer Hopf Algebra

This is the deepest and most surprising connection. It was a genuine mathematical discovery, not an observation, and it resolved a long-standing conceptual puzzle about the structure of renormalisation.

6.3.1 The Problem: Nested Subdivergences

In QFT, a Feynman diagram Γ may contain divergent subdiagrams $\gamma \subset \Gamma$. To renormalise Γ, one must subtract the subdivergences *before* subtracting the overall divergence. For deeply nested diagrams this leads to the complicated BPHZ forest formula (Bogoliubov–Parasiuk–Hepp–Zimmermann):

$$R(\Gamma) = \Gamma + \sum_{\text{forests } \mathcal{F}} \prod_{\gamma \in \mathcal{F}} (-C(\gamma)) \cdot \Gamma/\mathcal{F}, \tag{52}$$

where $C(\gamma)$ is the counterterm for subdiagram γ and $\Gamma/\mathcal{F}$ is the quotient graph. The sum is over all admissible forests of subdivergent subgraphs.

6.3.2 The Discovery: This Is a Hopf Algebra Antipode

Connes and Kreimer [14] showed that (52) is exactly the **antipode** S of a Hopf algebra H_R generated by *rooted trees*, where:

- Each 1PI Feynman diagram Γ maps to a rooted tree t_Γ whose nodes are the divergent subgraphs and whose root is Γ itself.
- The **coproduct** $\Delta(t) = \sum_c P_c(t) \otimes R_c(t)$ (summing over admissible cuts c) decomposes the tree into a subtree $P_c(t)$ (the subdivergence) and the quotient $R_c(t)$ (the remaining diagram after subtraction).
- The **antipode** $S(t) = -t - \sum_{c \neq \emptyset} S(P_c(t)) \cdot R_c(t)$ is the BPHZ counterterm.

6.3.3 The AC Identification

In AC language:

- The coproduct is an **admissible cut** of a rooted tree, which is exactly the decomposition used in the Lagrange analysis of tree GFs (the subtree above a cut corresponds to P_c; the base tree below corresponds to R_c).
- The antipode formula is the **Möbius inversion** / inclusion–exclusion that appears in the MSET and CYC constructions in AC.
- The **beta function** (the RG flow equation $\mu\partial_\mu g = \beta(g)$) is a **derivation** of the Hopf algebra H_R, and its perturbative expansion is a GF of rooted trees in the AC sense.

6.3.4 An Explicit Small Example

Consider the simplest non-trivial case: a one-loop self-energy subgraph γ inserted into a two-loop diagram Γ. The rooted tree is:

$$t_\Gamma = \begin{array}{c} \Gamma \\ | \\ \gamma \end{array}$$

The coproduct (the only non-trivial admissible cut is at the edge Γ—γ):

$$\Delta(t_\Gamma) = t_\Gamma \otimes 1 + 1 \otimes t_\Gamma + \gamma \otimes (\Gamma/\gamma).$$

The antipode:

$$S(t_\Gamma) = -t_\Gamma - S(\gamma) \cdot (\Gamma/\gamma) = -t_\Gamma + \gamma \cdot (\Gamma/\gamma).$$

This is exactly the BPHZ formula: subtract the subdivergence γ from Γ/γ, then subtract the result from Γ.

In AC terms: the GF for the class of rooted trees of depth 2 is $T_2(z) = z \cdot T_1(z)$ where $T_1(z) = z/(1-z)$, and the admissible cut is a coefficient extraction in the composition.

6.3.5 The Riemann–Hilbert Connection

Connes and Kreimer further showed [15] that the Birkhoff decomposition of a loop $\gamma : S^1 \to G$ in the group G of characters of H_R is the mathematical statement of renormalisation: the "positive part" of the decomposition is the renormalised amplitude and the "negative part" is the counterterm. This connects renormalisation to the **Riemann–Hilbert problem** — a deep link between physics and classical complex analysis.

Status

Rigorously established. Original papers: Connes & Kreimer [14, 15]. The combinatorial structure of the Hopf algebra (without the physics) was known earlier from Grossman–Larson (1989) in the context of numerical analysis (Runge–Kutta methods). The book by Yeats [16] is the most accessible treatment oriented toward combinatorialists. Manchon [18] provides a concise mathematical survey.

6.4 Connection 4: Factorial Divergence, Borel Summation, and the Transfer Theorem

6.4.1 The Problem: Perturbation Theory Always Diverges

The number of Feynman diagrams at loop order n grows as $n!$ for large n. This is a combinatorial fact: the number of ways to connect $2n$ legs in pairs (Wick contractions) is $(2n-1)!! \sim \sqrt{2}(2n/e)^n$, which grows faster than any exponential. Therefore the perturbative expansion:

$$Z(g) = \sum_{n=0}^{\infty} c_n g^n, \qquad c_n \sim K \cdot n! \cdot \alpha^{-n},$$

has *zero radius of convergence* for any $\alpha > 0$. This is not a defect of QFT; it is a fundamental feature.

6.4.2 The AC Explanation

The transfer theorem (29) immediately explains this. If $Z(g)$ has a singularity at $g_* = 1/\alpha$ of the form:

$$Z(g) \sim K(1 - g/g_*)^{-\beta}, \qquad \beta \in \mathbb{N},$$

then:

$$c_n = [g^n]Z(g) \sim \frac{K}{\Gamma(\beta)} \cdot n^{\beta-1} \cdot g_*^{-n}.$$

For a simple pole ($\beta = 1$): $c_n \sim K \cdot g_*^{-n}$, exponential growth. For the partition function of ϕ^4 theory, the singularity at finite g_* is associated with a **instanton** (a non-perturbative saddle point), and:

$$c_n \sim K \cdot n! \cdot g_*^{-n}.$$

The $n!$ growth corresponds to an **essential singularity** in the Borel plane, not a branch point or pole in g-space. In the Borel transform $\hat{Z}(t) = \sum_n (c_n/n!)t^n$, the $n!$ growth means the series converges (radius of convergence $|t| < g_*$), and $\hat{Z}(t)$ has a singularity at $t = g_*$. The Borel sum $Z(g) = \int_0^\infty e^{-t/g}\hat{Z}(t)\,\mathrm{d}t$ is obstructed by this singularity.

The Transfer Theorem in the Borel Plane

The type of singularity of $\hat{Z}(t)$ at $t = g_*$ determines the non-perturbative physics:

- **Simple pole** at $t = g_*$: $\hat{Z}(t) \sim (t - g_*)^{-1}$. Borel sum ambiguous; imaginary part $\sim e^{-g_*/g}$ is the instanton contribution.
- **Branch point** at $t = g_*$: $\hat{Z}(t) \sim (t - g_*)^{\alpha}$. Transfer theorem gives $c_n \sim n^{-\alpha-1}n!g_*^{-n}$. The $n!$ growth is present but with a power-law correction. These are **renormalons** (IR or UV), associated with factorial growth from perturbation theory that is not controlled by instantons.

6.4.3 Resurgence: The GF of All Corrections

The modern framework of **resurgence** (Écalle [19], developed for physics by Dunne–Ünsal [20]) promotes this to a full AC-style theory. The **trans-series**:

$$Z(g) \sim \sum_{k=0}^{\infty} e^{-kS_I/g} \sum_{n=0}^{\infty} c_n^{(k)} g^n$$

where S_I is the instanton action, packages the perturbative expansion around each non-perturbative sector into a separate GF. The Borel–Écalle summation procedure is the analytic continuation of the GF past its singularity, and the "Stokes phenomena" (ambiguity in the analytic continuation) cancel between different sectors — a non-trivial identity that is the resurgence statement.

In AC language: the full trans-series is a GF with a *multi-sheeted Riemann surface* as its domain, and resurgence is the statement that the GF has a consistent analytic structure across all sheets.

Example 6.3 (Airy Function as a Trans-Series)**.** The Airy equation $y'' = zy$ has a formal divergent solution:

$$\mathrm{Ai}(z) \sim \frac{e^{-2z^{3/2}/3}}{2\sqrt{\pi}z^{1/4}} \sum_{n=0}^{\infty} \frac{(-1)^n\Gamma(3n+1/2)}{54^n n!\Gamma(n+1/2)} z^{-3n/2}.$$

The $\Gamma(3n + 1/2)/n!$ coefficients grow as $\sim (3n)!/(n!)^2 \sim n! \cdot C^n$, showing factorial divergence. The AC viewpoint: $\mathrm{Ai}(z)$ is a Lagrange-type GF evaluated at the saddle point of a cubic integral, with a square-root branch point in the Borel plane at $t = 2/3$. The two Airy solutions Ai and Bi are the two branches of this GF, and resurgence explains how they are related via the Stokes phenomenon.

Status

The factorial growth / Borel singularity connection is well-known in the resurgence literature [19, 20] but phrased differently from AC. The AC transfer theorem is not cited in the resurgence literature. The two communities (resurgence and AC) work with the same Borel transform but have not explicitly bridged their frameworks. This appears to be a genuine gap in the literature.

6.5 The Status Map: What Is Established and What Is New

The following table summarises the state of the art for each connection.

Connection	**Status**	**Key References**
$e^{C(z)} = Z$; symmetry factors from EGF	Classical folklore	Flajolet–Sedgewick [11], Zinn-Justin [21]
Dyson–Schwinger = Lagrange equation	Known implicitly	Foissy [17], Yeats [16]
Lagrange inversion generates loop expansion	Not stated explicitly	–
Landau pole = Lagrange branch point	Not stated explicitly	–
Connes–Kreimer: BPHZ = Hopf antipode	Rigorously proved	Connes–Kreimer [14, 15]
Beta function = derivation of H_R	Rigorously proved	Connes–Kreimer [15]
Renormalisation = Riemann–Hilbert problem	Rigorously proved	Connes–Kreimer [15]
Factorial growth ⇔ Borel singularity	Known (separately)	Écalle [19], Dunne–Ünsal [20]
Borel singularity = AC transfer theorem	Not bridged	–
Trans-series = multi-sheet GF	Implicit in resurgence	Écalle [19]
SIR orbit = singular locus of epidemic GF	Follows from Kendall [3]	Not stated explicitly
Doi–Peliti saddle = AC branch point	Not in literature	–
SIR / Doi–Peliti / AC triangle	Not synthesised	This document

6.6 The Grand Summary: One Table

All four connections can be summarised as a dictionary between AC and QFT:

Analytic Combinatorics	**Quantum Field Theory**
EGF of connected structures $\hat{C}(z)$	Free energy $F = -\log Z$
$e^{\hat{C}(z)}$ exponential formula	$Z = e^{\text{connected diagrams}}$
Symmetry factor $= 1/\lvert\text{Aut}\rvert$	EGF overcounting $1/k!$ in SET
Lagrange equation $T = z\phi(T)$	Dyson–Schwinger equation
Lagrange inversion coefficients	Feynman diagram enumeration
Branch point of Lagrange GF	Landau pole / non-pert. scale
Rooted tree Hopf algebra coproduct	BPHZ subdivergence subtraction
Hopf antipode	Renormalisation counterterms
Möbius inversion on forests	Forest formula (BPHZ)
GF singularity type	Universality class / RG fixed point
Borel transform singularity	Instanton / renormalon
Transfer theorem: $n^{-\alpha-1} \cdot A^n$	Perturbative coefficient growth
Multi-sheet Riemann surface	Trans-series / resurgence

Why This Matters

The table above is not merely a curiosity. It suggests that the *combinatorial structure* of a QFT — the grammar by which Feynman diagrams are built — is the primary object, and the analytic structure (propagators, amplitudes, renormalisation) is a *representation* of that combinatorial object. This is the philosophy behind the Connes–Kreimer programme, and it has led to concrete new results: the Birkhoff decomposition of renormalisation, the connection to the Riemann–Hilbert problem, and new methods for computing multi-loop amplitudes.

For the stochastic processes studied in this document (SIR, $A + A \to \emptyset$), the same philosophy applies: the epidemic tree, the first-passage excursion, and the Doi–Peliti path integral are different representations of the same combinatorial grammar, and their common singularity structure is the physical content.

A Table of Key Results

A.1 Generating Function Identities

Object	GF	Notes
Catalan numbers $C_n = \binom{2n}{n}/(n+1)$	$C(z) = (1-\sqrt{1-4z})/2z$	Binary trees
Borel distribution	$T(z) = ze^{R_0(T-1)}$	SIR final size (Lagrange)
First-passage (1D)	$F(z) = (1-\sqrt{1-z^2})/z$	Ballot paths
Poisson PGF	$g(s) = e^{\lambda(s-1)}$	Galton–Watson offspring
Geometric GF	$1/(1-z)$	Sequences of atoms

A.2 Transfer Theorem Reference

Singularity	Local form near z_*	$[z^n]f$ behaviour	Example
Simple pole	$(1-z/z_*)^{-1}$	$\sim z_*^{-n}$	Fibonacci
k-th pole	$(1-z/z_*)^{-k}$	$\sim n^{k-1}z_*^{-n}$	Path lengths
Sq. root +	$(1-z/z_*)^{1/2}$	$\sim n^{-3/2}z_*^{-n}$	SIR, A+A
Sq. root −	$(1-z/z_*)^{-1/2}$	$\sim n^{-1/2}z_*^{-n}$	Simple RW
Log	$-\log(1-z/z_*)$	$\sim n^{-1}z_*^{-n}$	Cycles

A.3 Doi–Peliti Reaction Dictionary

Reaction	Rate	Liouvillian $\mathcal{H}$
$A \to \emptyset$	δ	$\delta(1-a^\dagger)a$
$\emptyset \to A$	σ	$\sigma(a^\dagger - 1)$
$A \to 2A$	λ	$\lambda(a^{\dagger 2} - a^\dagger)a$
$2A \to \emptyset$	μ	$\mu(1-a^{\dagger 2})a^2$
$2A \to A$	κ	$\kappa(a^\dagger - a^{\dagger 2})a^2$
$A+B \to \emptyset$	ν	$\nu(1-a^\dagger{}_A a^\dagger{}_B)a_A b_B$

The general rule: reaction $kA \to \ell A$ at rate r gives $\mathcal{H} = r(a^{\dagger\ell} - a^{\dagger k})a^k$.

A.4 Critical Dimensions for Reaction–Diffusion

Reaction	Upper d_c	Exponent α ($d < d_c$)	Mean-field ($d > d_c$)
$A+A \to \emptyset$	2	$d/2$	1
$A+A \to A$	2	$d/2$	1
$A+B \to \emptyset$	4	$d/4$	1/2
$3A \to \emptyset$	1	$d/2$	1

A.5 The Weyl Algebra and the GF Isomorphism

The **Weyl algebra** W_1 is the unital associative algebra generated by p and q with relation $[p,q] = 1$ (or equivalently, $pq - qp = 1$). Under the representation $p = \partial_z$, $q = z$ (multiplication), W_1 acts on $\mathbb{C}[[z]]$. This is the unique irreducible representation of W_1 on a space of formal power series (Stone–von Neumann theorem in the algebraic sense).

Doi's identification: $a \equiv \partial_z$, $a^\dagger \equiv z$ gives an isomorphism:

- Fock space $\mathcal{F} \cong \mathbb{C}[[z]]$ as W_1-modules.
- The state $|\psi\rangle = \sum_n P_n|n\rangle$ corresponds to $G(z) = \sum_n P_n z^n$.
- The Liouvillian $\mathcal{H}(a, a^\dagger)$ corresponds to $\mathcal{H}(\partial_z, z)$.
- The evolution equation $\partial_t|\psi\rangle = \mathcal{H}|\psi\rangle$ becomes the GF PDE $\partial_t G = \mathcal{H}(\partial_z, z)G$.

A.6 The Hopf Algebra Connection

The coproduct of W_1 (describing how two particles interact) is:

$$\Delta(a) = a \otimes 1 + 1 \otimes a, \qquad \Delta(a^\dagger) = a^\dagger \otimes 1 + 1 \otimes a^\dagger.$$

The combinatorial counterpart: the coproduct of the Connes–Kreimer Hopf algebra of rooted trees decomposes a tree into subtree/quotient pairs, which is precisely the admissible cut operation. Renormalisation (the antipode of the Hopf algebra) corresponds to the inclusion–exclusion in Flajolet's symbolic method for the MSET and CYC constructions. This is why the Feynman diagram expansion of the Doi–Peliti theory and the AC generating function equations are algebraically the same object at different levels of abstraction.

B Literature Guide

B.1 Stochastic SIR

- **Kermack & McKendrick (1927)** [1]: the original deterministic SIR paper. Derives the epidemic threshold and conservation law.
- **Whittle (1955)** [2]: first use of the PGF for the stochastic SIR. Derives the minor-outbreak probability.
- **Kendall (1956)** [3]: explicit identification of the characteristic equations of the PGF PDE with the deterministic SIR.
- **Chalub & Souza (2011)** [4]: modern PDE formulation. Proves that the SIR ODEs are the characteristics and derives large-population limits rigorously.

B.2 Doi–Peliti

- **Doi (1976a,b)** [5, 6]: two papers introducing the second-quantisation formalism for reaction–diffusion master equations.
- **Peliti (1985)** [7]: coherent-state path integral for the Doi formalism. Introduces the $\phi, \tilde{\phi}$ field doubling.
- **Lee (1994)** [8]: RG calculation for $kA \to \emptyset$. Proves that the density exponent is exact.
- **Täuber, Howard & Vollmayr-Lee (2005)** [9]: the comprehensive review. Covers $kA \to \ell A$, $A + B \to \emptyset$, disorder, and absorbing-state transitions. Best entry point.
- **Täuber (2014)** [10]: textbook treatment, Chapter 9 on reaction–diffusion.

B.3 Analytic Combinatorics

- **Flajolet & Sedgewick (2009)** [11]: the canonical reference. Parts A–B cover the symbolic method; Part C covers singularity analysis and transfer theorems. Freely available online.
- **Wilf (2006)** [12]: *generatingfunctionology*, a more accessible introduction to generating functions in combinatorics.
- **Pemantle & Wilson (2013)** [13]: multivariate analytic combinatorics; extends the singularity approach to GFs in multiple variables.

B.4 AC, Feynman Diagrams, and Renormalisation

- **Connes & Kreimer (1998)** [14]: the discovery that the BPHZ forest formula is the antipode of a Hopf algebra of rooted trees. The paper that started the programme. Dense but repays careful reading.
- **Connes & Kreimer (2000)** [15]: the Riemann–Hilbert problem formulation of renormalisation. Shows the beta function is a derivation of the Hopf algebra and that the Birkhoff decomposition encodes counterterms.
- **Yeats (2017)** [16]: *A Combinatorial Perspective on Quantum Field Theory*. The best entry point for a mathematician. Explicitly begins from perturbative expansions as generating functions, introduces the Connes–Kreimer algebra, and works through Dyson–Schwinger equations as combinatorial fixed-point equations. The table of contents mirrors the structure of Section 6 of this document.
- **Foissy (2007)** [17]: combinatorial Dyson–Schwinger equations as systems of equations in the Hopf algebra of planar trees. This is the closest existing work to the "Lagrange equation = DSE" identification in Section 6.2.
- **Manchon (2004)** [18]: a concise mathematical survey of Hopf algebras in renormalisation, including the connection to Möbius inversion and the BPHZ formula.
- **Kreimer (2000)**: *Combinatorics of (perturbative) Quantum Field Theory*, arXiv:hep-th/0010059. A readable overview of how Feynman graphs map to rooted trees and how the Hopf algebra coproduct encodes subdivergences.

B.5 Resurgence and Borel Summation

- **Écalle (1981)** [19]: the original mathematical framework of resurgence and analysable functions. Very technical. The three-volume work that introduced trans-series.
- **Dunne & Ünsal (2014)** [20]: the clearest modern physics treatment of resurgence with explicit QFT examples. Introduces the connection between IR renormalons (Borel singularities) and non-perturbative contributions.
- **Zinn-Justin (2002)** [21]: standard QFT textbook with good treatment of the large-order behaviour of perturbation theory, instantons, and Borel summation. Chapter 37–40 are directly relevant.

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